Multiple Access Control (MAC) Protocols

Coverage: Random Access (Pure & Slotted ALOHA, CSMA, CSMA/CD, CSMA/CA), Controlled Access (Polling, Token Ring), CDMA

Prepared by: Prof. Anju Markose | Assistant Professor, Dept. of Computer Science & Engineering | VJCET

1. ALOHA Protocols

• Pure ALOHA: S = G * e^(-2G), Max 18.4%. Slotted ALOHA: S = G * e^(-G), Max 36.8%.

2. CSMA/CD

• Carrier Sense + Collision Detection. Minimum frame size L_min = 2 * t_prop * R.

• Binary Exponential Backoff: Wait r * 512 bit times where r in [0, 2^k - 1].

■ Worked Problem: CSMA/CD Minimum Frame Size

• Given R = 1 Gbps, d = 1 km, s = 2*10^8 m/s -> t_prop = 5 microseconds. L_min = 10^-5 * 10^9 = 1250 Bytes.

■ KTU University High-Yield Exam Questions

• Q: Derive throughput for Pure ALOHA and Slotted ALOHA and explain CSMA/CD. (8 Marks)